Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/23/2018
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Fundamental Research on International Standard of Fire Safety in Space - Subteam 2: Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	510 642-6554
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for Fundamental Research on an International Standard of Fire Safety in Space
Start Date:	03/06/2014	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:	ISS		
Flight Assignment:	NOTE: New end date is 12/31/2018 per NSSC information (Ed., 10/16/18) NOTE: New end date is 9/05/2018 per NSSC information (Ed., 4/12/18)		
Key Personnel Changes/Previous PI:	Prof. Carlos Fernandez-Pello is U.S. Co-Investigator on Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA Principal Investigator is Prof. Osamu Fujita, Hokkaido University.		
COI Name (Institution):			
Grant/Contract No.:	NNX14AF01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Prof. Fernandez-Pello's role as U.S. Co-Investigator for the Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA International Announcement of Opportunity (AO) to fund experiments to be conducted aboard the Japanese Experiment Module, Kibo, 2012. The objective of the proposed research program is to continue the experimental study of the flammability of wire materials in space exploration atmospheres and associated computational/theoretical tools to aid interpretation of test results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in spacecraft allows us to accurately elucidate the effect of the environment parameters on the ignition and flame spread over combustible material, and through them their potential fire hazard. Particularly important is the determination of the Limiting Oxygen Concentration (LOC) on flame extinction under spacecraft environments. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. Another important aspect of the research is the effect of melting and dripping of plastic insulation in normal gravity in comparison with microgravity. The results are relevant because dripping will not occur in microgravity and consequently could impact their burning and methodology to screen. The investigation and results have also benefits for terrestrial fire safety by providing further information about the flammability of materials under a variety of environments.
Task Progress:	The research has one primary goal: to understand the changes in electrical wire flammability in the environment expected in space-based facilities. An aspect of the problem is that if the insulation material melts during the wire burning, major differences in flammability may occur between normal and microgravity, because the molten material will not drip in microgravity. The overall project experimental methodology consists of obtaining flame spread flammability characteristics of electrical wires and wire insulation materials (primarily those being studied in Hokkaido University) under external radiation. Experiments were conducted primarily in normal gravity to observe the effect of melting and dripping on the flame spread over wire insulation. Effects, such as wire orientation, external radiation and oxygen concentration, were analyzed for different wire geometries. A few tests were conducted in microgravity at Earth atmospheres at the 5 seconds NASA Glenn Research Center (GRC) drop tower and in a parabolic flight at Hokkaido University. The results of the microgravity tests are currently being analyzed.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Abstracts for Journals and Proceedings	Kobayashi Y, Konno Y, Huang X, Tsue M, Fujita O, Fernandez-Pello C. "Flame Spread and Dripping Behaviors in Horizontal and Vertical Wires." Presented at the 10th U.S. National Combustion Meeting, College Park, MD, April 23-26, 2017. 10th U.S. National Combustion Meeting, College Park, MD, April 23-26, 2017. , Apr-2017
Articles in Peer-reviewed Journals	Kobayashi Y, Huang X, Nakaya S, Tsue M, Fernandez-Pello C. "Flame spread over horizontal and vertical wires: The role of dripping and core." Fire Safety Journal. 2017 Jul;9:112-22. https://doi.org/10.1016/j.firesaf.2017.03.047 , Jul-2017
Articles in Peer-reviewed Journals	Kobayashi Y, Konno Y, Huang X, Nakaya S, Tsue M, Hashimoto N, Fujita O, Fernandez-Pello C. "Effect of insulation melting and dripping on opposed flame spread over laboratory simulated electrical wires." Fire Safety Journal. 2018 Jan;95:1-10. https://doi.org/10.1016/j.firesaf.2017.10.006 , Jan-2018